

TIMIRYAZEVSIIY, Russian Federation

WMO: 319610

Lat: 43.8571N

Lon: 131.9530E

Elev: 119

StdP: 14.63

Time Zone: 10.00 (E10)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.5	-17.9	-27.9	1.2	-19.8	-23.7	1.5	-16.2	19.9	8.6	18.2	13.9	2.3	50	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.8	85.4	72.6	82.3	71.3	79.3	69.8	75.3	82.1	73.6	79.0	71.9	77.0	5.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	124.4	78.7	71.8	117.8	77.3	70.1	111.4	75.2	39.0	82.0	37.4	79.2	35.9	77.2	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	16.7	14.9	-25.9	91.4	6.3	2.6	-30.5	93.3	-34.2	94.8	-37.8	96.3	-42.4	98.2		
(5)			-26.2	77.7	6.2	1.2	-30.7	78.6	-34.3	79.3	-37.8	80.0	-42.3	80.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.6	1.9	11.0	28.5	43.6	55.8	62.9	70.7	71.1	61.0	45.9	26.3	6.9		
	DBStd	25.28	9.20	10.00	10.36	6.56	5.83	5.48	4.83	4.83	6.45	7.84	11.62	10.16		
	HDD50	5706	1491	1093	667	210	16	0	0	0	5	176	713	1335		
	HDD65	9356	1956	1513	1131	642	288	101	10	10	150	593	1162	1800		
	CDD50	2280	0	0	1	18	196	387	640	655	334	48	1	0		
	CDD65	455	0	0	0	0	4	38	185	199	29	0	0	0		
	CDH74	2341	0	0	0	3	80	246	919	984	109	0	0	0		
	CDH80	528	0	0	0	0	13	64	241	201	9	0	0	0		
	WSAvg	6.6	4.8	5.8	7.3	8.5	8.8	8.1	7.2	5.7	5.4	6.4	6.2	5.1		
	PrecAvg	25.2	0.3	0.5	0.6	1.6	2.1	3.3	3.3	5.0	4.5	2.2	1.1	0.5		
	PrecMax	39.8	1.0	2.3	2.4	4.8	4.2	12.4	7.4	14.2	10.5	6.7	3.4	2.0		
(15) Precipitation	PrecMin	16.6	0.0	0.0	0.0	0.3	0.1	0.9	0.3	0.0	0.4	0.3	0.0	0.0		
	PrecStd	6.3	0.3	0.5	0.5	1.1	1.1	2.1	1.7	2.9	2.6	1.5	0.8	0.5		
	DB	31.6	40.7	59.9	71.8	80.4	86.4	89.4	88.2	81.1	69.6	57.3	39.6			
	MCWB	29.0	31.7	45.4	52.2	62.0	70.3	76.0	74.2	66.6	58.9	49.5	36.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	27.3	36.8	52.4	64.8	75.8	80.5	85.7	84.9	76.9	66.3	52.4	33.2			
	2% MCWB	24.4	33.2	40.5	48.4	58.2	67.2	73.6	72.8	65.6	55.6	45.6	30.2			
	5% DB	22.2	33.6	47.1	60.4	72.0	76.0	82.7	82.0	74.2	62.9	48.1	28.8			
	5% MCWB	18.6	30.2	37.3	46.1	56.4	65.1	72.7	71.6	64.5	54.6	42.8	25.7			
	10% DB	17.6	28.6	42.8	56.1	67.9	72.2	79.1	79.4	71.6	60.0	43.8	24.1			
	10% MCWB	14.9	24.2	35.4	43.8	55.0	64.0	71.0	70.6	63.4	52.9	39.4	21.2			
	0.4% WB	30.1	36.5	46.1	53.3	63.3	72.4	78.1	76.5	71.5	63.5	51.2	36.5			
	0.4% MCDB	31.6	37.7	58.6	68.3	78.3	83.2	85.8	83.4	75.0	67.3	53.9	38.6			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	24.8	33.8	41.7	50.1	60.0	68.8	75.8	75.4	68.9	59.4	46.9	30.8			
	2% MCDB	27.1	35.7	50.1	62.0	72.1	78.0	83.2	81.5	73.4	63.2	50.8	33.0			
	5% WB	18.7	29.5	38.6	47.7	58.1	66.5	73.8	73.9	66.8	57.0	43.4	25.8			
	5% MCDB	21.9	34.3	46.0	56.5	68.2	73.5	80.1	78.9	71.9	61.6	47.6	28.6			
	10% WB	14.8	24.5	35.9	45.4	56.3	64.7	71.9	72.6	65.1	54.0	39.5	21.3			
	10% MCDB	17.6	28.1	41.9	53.9	65.2	70.7	77.5	77.1	70.2	58.3	43.6	24.1			
	MDBR	23.9	24.0	19.4	19.6	18.4	14.1	13.0	13.8	19.4	21.6	18.2	21.0			
	5% DB	25.8	22.0	25.2	28.5	27.5	22.2	19.3	17.9	20.8	24.5	22.0	21.2			
(36) Mean Daily Temperature Range	MCWBR	23.0	17.5	14.6	14.7	12.1	10.6	8.8	7.4	10.1	14.8	15.4	18.5			
	MCDBR	25.6	20.3	22.3	24.1	22.5	18.7	16.9	13.9	15.3	16.8	19.3	21.0			
	MCWBR	23.2	16.6	13.4	13.5	11.1	9.6	8.4	6.9	8.4	12.1	15.5	19.0			
	MCDBR	23.2	16.6	13.4	13.5	11.1	9.6	8.4	6.9	8.4	12.1	15.5	19.0			
(40) Clear-Sky Solar Irradiance	taub	0.328	0.385	0.474	0.580	0.594	0.568	0.534	0.488	0.440	0.457	0.398	0.345			
	taud	2.077	1.920	1.807	1.663	1.699	1.840	1.989	2.113	2.155	1.966	2.024	2.053			
	Ebn at Noon	240	243	236	222	224	231	238	245	247	216	209	218			
	Edh at Noon	33	46	60	75	74	64	55	47	42	44	35	31			
(44) All-Sky Solar Radiation	RadAvg	705	1032	1351	1460	1605	1671	1500	1446	1332	979	679	576			
	RadStd	44	56	95	135	143	201	179	137	113	75	55	33			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page